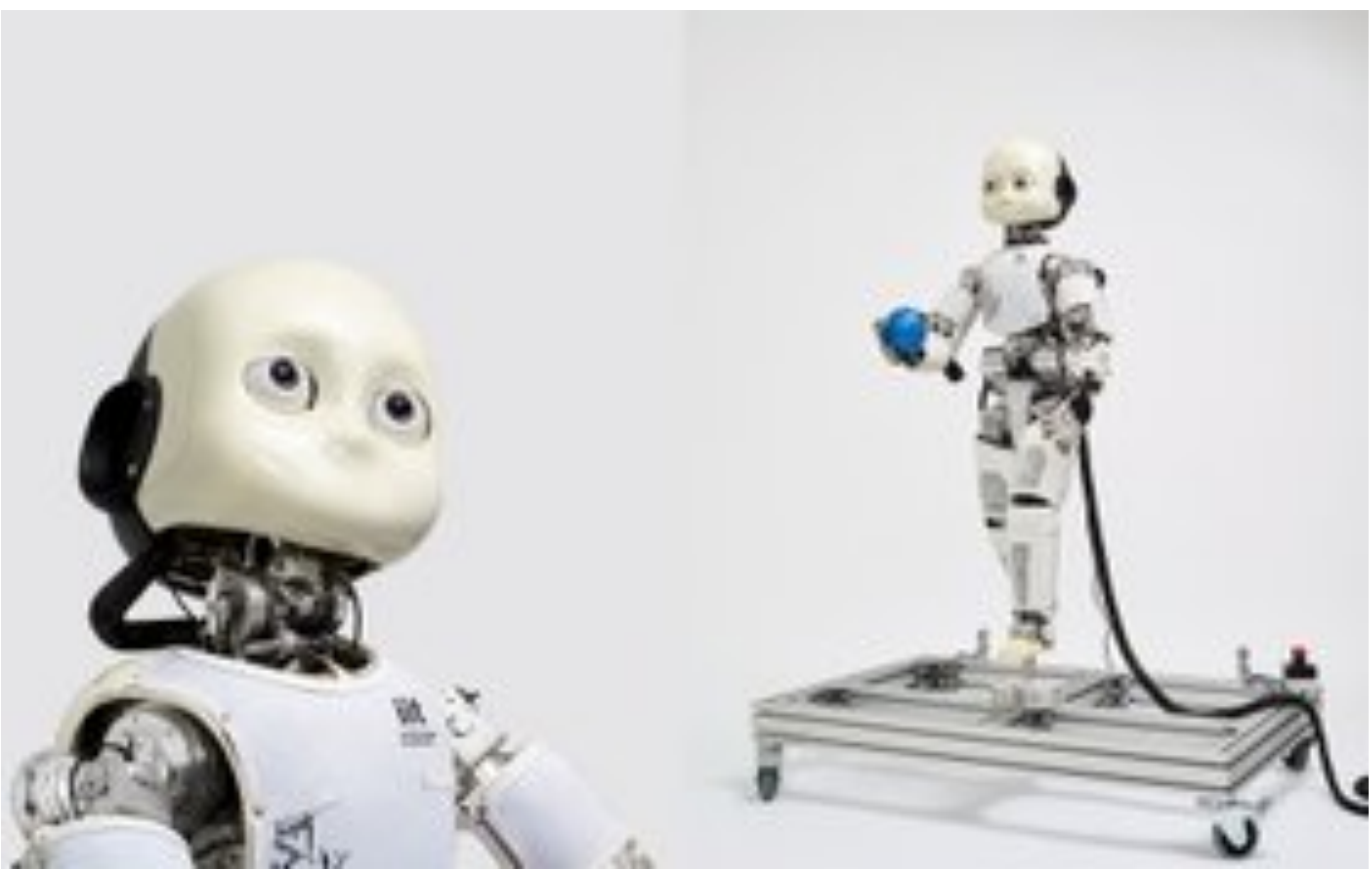
COMPUTING

Intelligent Machines That Learn Like Children

Machines that learn like children provide deep insights into how the mind and body act together to bootstrap knowledge and skills

By Diana Kwon | Scientific American March 2018 Issue



Credit: Sun Lee

IN BRIEF

- **Infants learn** autonomously, by experimenting with their bodies and playing with objects.
- **Roboticists** are programming androids with algorithms that enable them to learn like children.
- **Studies with such machines** are transforming robotics and providing insights into child development.

Deon, a fictional engineer in the 2015 sci-fi film *Chappie*, wants to create a machine that can think and feel. To this end, he writes an artificial-intelligence program that can learn like a child. Deon's test subject, Chappie, starts off with a relatively blank mental slate. By simply observing and experimenting with his surroundings, he acquires general knowledge, language and complex skills—a task that eludes even the most advanced AI systems we have today.

To be sure, certain machines already exceed human abilities for specific tasks, such as playing games like Jeopardy, chess and the Chinese board game Go. In October 2017 British company DeepMind unveiled AlphaGo Zero, the latest version of its AI system for playing Go. Unlike its predecessor AlphaGo, which had mastered the game by mining vast numbers of human-played games, this version accumulated experience autonomously, by competing against itself. Despite its remarkable achievement, AlphaGo Zero is limited to learning a game with clear rules—and it needed to play millions of times to gain its superhuman skill.

In contrast, from early infancy onward our offspring develop by exploring their surroundings and experimenting with movement and speech. They collect data themselves, adapt to new situations and transfer expertise across domains.

Since the beginning of the 21st century, roboticists, neuroscientists and psychologists have been exploring ways to build machines that mimic such spontaneous development. Their collaborations have resulted in androids that can move objects, acquire basic vocabulary and numerical abilities, and even show signs of social behavior. At the same time, these AI systems are helping psychologists understand how infants learn.

PREDICTION MACHINE

Our brains are constantly trying to predict the future—and updating their expectations to match reality. Say you encounter your neighbor's cat for the first time. Knowing your own gregarious puppy, you expect that the cat will also enjoy your caresses. When you reach over to pet the creature, however, it scratches you. You update your theory about cuddly-looking animals—surmising, perhaps, that the kitty will be friendlier if you bring it a treat. With goodies in hand, the cat indeed lets you stroke it without inflicting wounds. Next time you encounter a furry feline, you offer a tuna tidbit before trying to touch it.

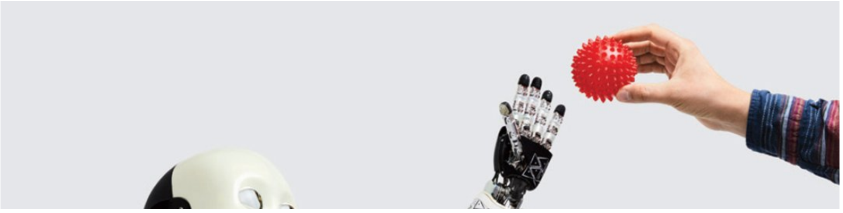
In this manner, the higher processing centers in the brain continually refine their internal models according to the signals received from the sensory organs. Take our visual systems, which are highly complex. The nerve cells in the eye process basic features of an image before transferring this information to higher-level regions that interpret the overall meaning of a scene. Intriguingly, neural connections also run in the other direction: from high-level processing centers, such as areas in the parietal or temporal cortices, to low-level ones such as the primary visual cortex and the lateral geniculate nucleus [*see graphic below*]. Some neuroscientists believe that these "downward" connections carry the brain's predictions to lower levels, influencing what we see.

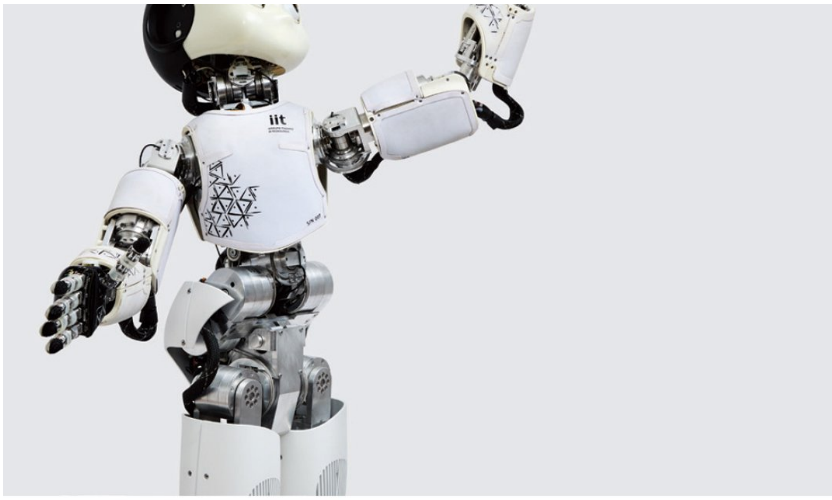
Crucially, the downward signals from the higher levels of the brain continually interact with the "upward" signals from the senses, generating a prediction error: the difference between what we expect and what we experience. A signal conveying this discrepancy returns to the higher levels, helping to refine internal models and generating fresh guesses, in an unending loop. "The prediction error signal drives the system toward estimates of what's really out there," says Rajesh P. N. Rao, a computational neuroscientist at the University of Washington.

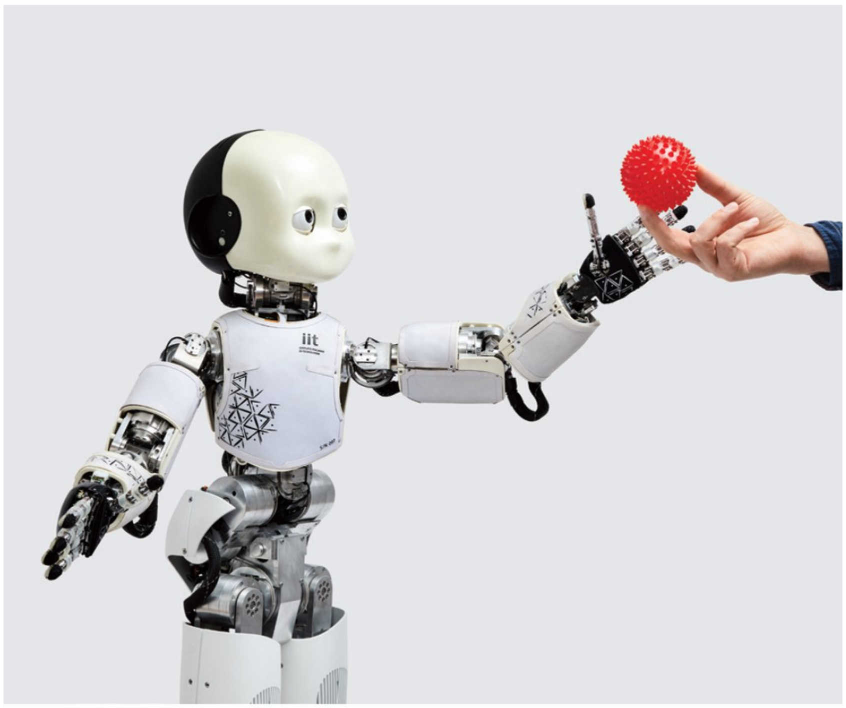
While Rao was a doctoral student at the University of Rochester, he and his supervisor, computational neuroscientist Dana H. Ballard, now at the University of Texas at Austin, became the first to test such predictive coding in an artificial neural network. (A class of computer algorithms modeled on the human brain, neural networks incrementally adapt internal parameters to generate the required output from a given input.) In this computational experiment, published in 1999 in *Nature Neuroscience*, the researchers simulated neuronal connections in the visual cortex—complete with downward connections carrying forecasts and upward connections bringing sensory signals from the outside world. After training the network using pictures of nature, they found that it could learn to recognize key features of an image, such as a zebra's stripes.

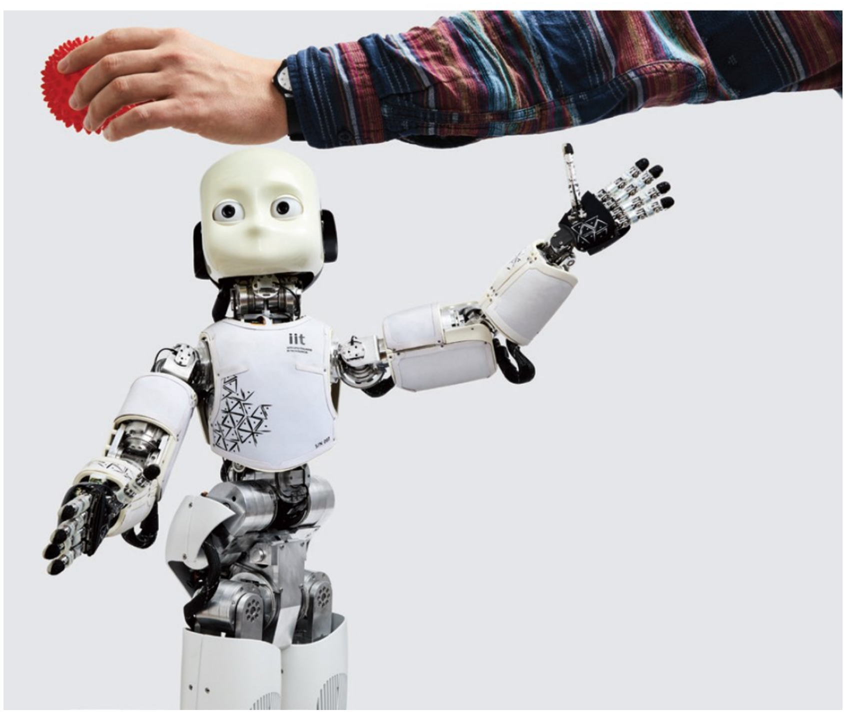
COUNTING WITH FINGERS

A fundamental difference between us and many present-day AI systems is that we possess bodies that we can use to move about and act in the world. Babies and toddlers develop by testing the movements of their arms, legs, fingers and toes and examining everything within reach. They autonomously learn how to walk, talk, and recognize objects and people. How youngsters are able to do all this with very little guidance is a key area of investigation for both developmental psychologists and roboticists. Their collaborations are leading to surprising insights—in both fields.









ICUB, an android being studied at the University of Plymouth in England, can learn new words, such as "ball," more easily if the experimenter consistently places the object at the same location while naming it. *Sun Lee*

In a series of pioneering experiments starting in the late 1990s, roboticist Jun Tani, then at Sony Computer Science Laboratories, and others developed a prediction-based neural network for learning basic movements and tested how well these algorithms worked in robots. The machines, they discovered, could attain elementary skills such as navigating simple environments, imitating hand movements, and following basic verbal commands like "point" and "hit."

More recently, roboticist Angelo Cangelosi of the University of Plymouth in England and Linda B. Smith, a developmental psychologist at Indiana University Bloomington, have demonstrated how crucial the body is for procuring knowledge. "The shape of the [robot's] body, and the kinds of things it can do, influences the experiences it has and what it can learn from," Smith says. One of the scientists' main test subjects is iCub, a three-foot-tall humanoid robot built by a team at the Italian Institute of Technology for research purposes. It comes with no preprogrammed functions, allowing scientists to implement algorithms specific to their experiments.

In a 2015 study, Cangelosi, Smith and their colleagues endowed an iCub with a neural network that gave it the ability to learn simple associations and found that it acquired new words more easily when objects' names were consistently linked with specific bodily positions. The experimenters repeatedly placed either a ball or a cup to the left or right of the android, so that it would associate the objects with the movements required to look at it, such as tilting its head. Then they paired this action with the items' names. The robot was better able to learn these basic words if the corresponding objects appeared in one specific location rather than in multiple spots.

Interestingly, when the investigators repeated the experiment with 16-month-old toddlers, they found similar results: relating objects to particular postures helped small children learn word associations. Cangelosi's laboratory is developing this technique to teach robots more abstract words such as "this" or "that," which are not linked to specific things.

Using the body can also help children and robots gain basic numerical skills. Studies show, for instance, that youngsters who have difficulty mentally representing their fingers also tend to have weaker arithmetic abilities. In a 2014 study, Cangelosi and his team discovered that when the robots were taught to count with their fingers, their neural networks represented numbers more accurately than when they were taught using only the numbers' names.

CURIOSITY ENGINES

Novelty also helps children learn. In a 2015 *Science* paper, researchers at Johns Hopkins University reported that when infants encounter the unknown, such as a solid object that appears to move through a wall, they explore their violated expectations. In prosaic terms, their in-built drive to reduce prediction errors aids their development.

Pierre-Yves Oudeyer, a roboticist at INRIA, the French national institute for computer science, believes that the learning process is more complex. He holds that kids actively, and with surprising sophistication, seek out those objects in their environment that provide greater opportunities to learn. A toddler, for example, will likely choose to play with a toy car rather than with a 100-piece jigsaw puzzle—arguably because her level of knowledge will allow her to generate more testable hypotheses about the former.

To test this theory, Oudeyer and his colleagues endowed robotic systems with a feature they call intrinsic motivation, in which a decrease in prediction error yields a reward. (For an intelligent machine, a reward can correspond to a numerical quantity that it has been programmed to maximize through its actions.) This mechanism enabled a Sony AIBO robot, a small, puppylike machine with basic sensory and motor abilities, to autonomously seek out tasks with the greatest potential for learning. The robotic puppies were able to acquire basic skills, such as grasping objects and interacting vocally with another robot, without having to be programmed to achieve these specific ends. This outcome, Oudeyer explains, is "a side effect of the robot exploring the world, driven by the motivation to improve its predictions."

Remarkably, even though the robots went through similar stages of training, chance played a role in what they learned. Some explored a bit less, others a bit more—and they ended up knowing different things. To Oudeyer, these varied outcomes suggest that even with identical programming and a similar educational environment, robots may attain different skill levels—much like what happens in a typical classroom.

More recently, Oudeyer's group used computational simulations to show that robotic vocal tracts equipped with these predictive algorithms (and the proper hardware) could also learn basic elements of language. He is now collaborating with Jacqueline Gottlieb, a cognitive neuroscientist at Columbia University, to investigate whether such prediction-driven intrinsic motivation underlies the neurobiology of human curiosity as well. Probing these models further, he says, could help psychologists understand what happens in the brains of children with developmental disabilities and disorders.

ALTRUISTIC ANDROIDS

Our brains also try to forecast the future when we interact with others: we constantly attempt to deduce people's intentions and anticipate what they might say next. Intriguingly, the drive to reduce prediction errors can, in and of itself, induce elementary social behavior, as roboticist Yukie Nagai and her colleagues demonstrated in 2016 at Osaka University in Japan.

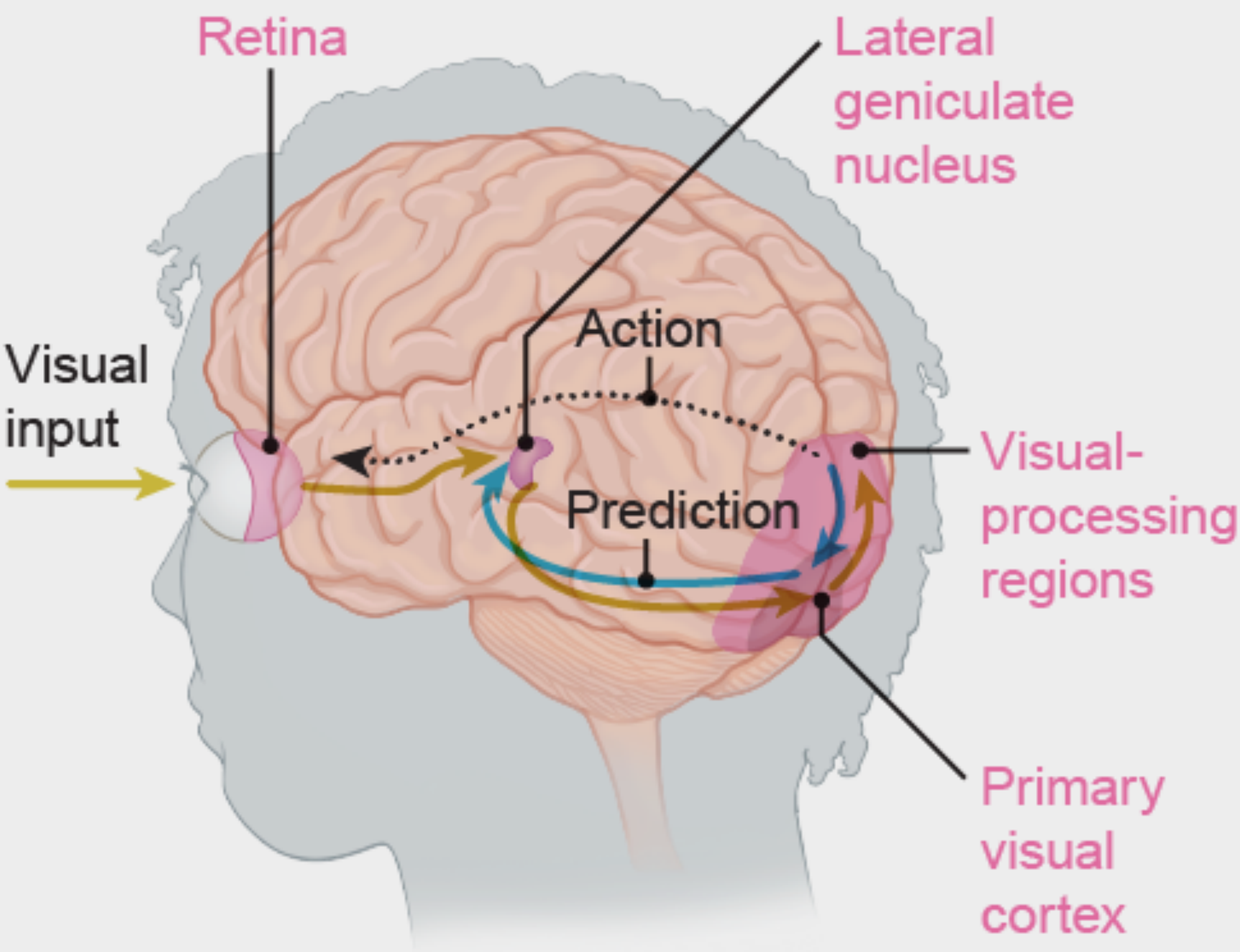
The researchers found that even when iCub was not programmed with an intrinsic ability to socialize, the motivation to reduce prediction errors alone led it to behave in a helpful way. For example, after the android was taught to push a toy truck, it might observe an experimenter failing to complete that same action. Often it would move the object to the right place—simply to increase the certainty of the truck being at a given location. Young children might develop in a similar way, believes Nagai, who is currently at the National Institute of Information and Communications Technology in Japan. “The infant doesn't need to have the intention to help other persons,” she argues: the motivation to minimize prediction error can alone initiate elementary social abilities.

Predictive Brain

Our minds are prediction machines, using prior experience and knowledge to make sense of the deluge of information coming from our surroundings. Many neuroscientists and psychologists believe that nearly everything we do—perception, action and learning—relies on making and updating expectations.

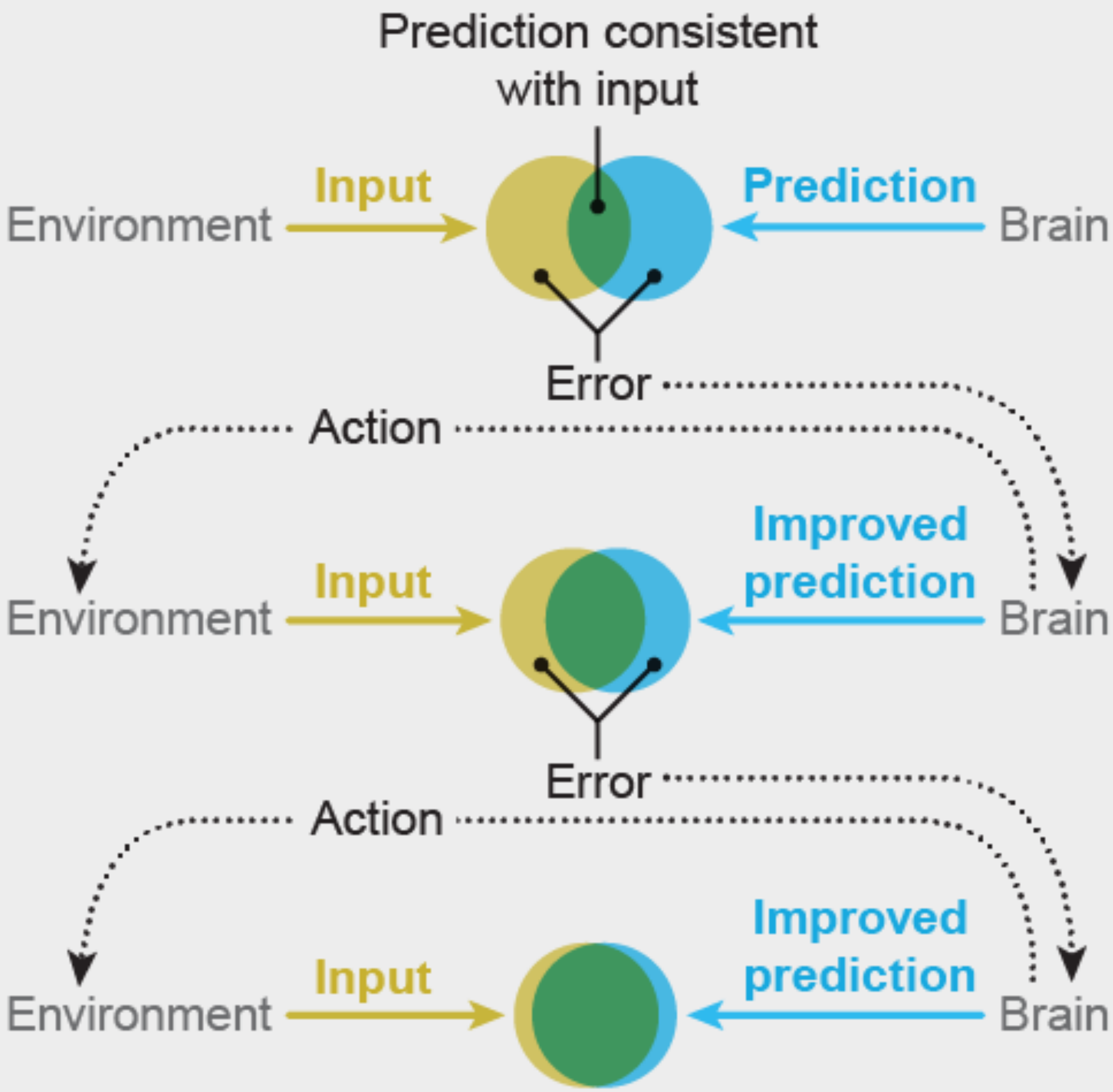
Visual Processing

The brain’s anatomy supports the idea of predictive processing. The visual cortex, for example, receives inputs from the eye, but connections also run in the other direction. Neuroscientists believe that these “downward” connections, from higher levels of the "downward" connections, from higher levels of the brain to the lower (such as the primary visual cortex and the lateral geniculate nucleus), carry predictions. These meet with the sensory input to generate a prediction error: the difference between what you expect and what you see. A signal coding this discrepancy returns to the higher levels of the brain. Other downward signals send commands to move the eye muscles, adjusting what we see.



Cascade of Predictions

When the brain generates a prediction error, it uses this information to update its expectations and select actions that will help resolve the discrepancy between beliefs and reality. For example, if an individual cannot determine what an object is simply by looking at it, the brain might send a command to pick up the item and examine it to gather more information.



Credit: Mesa Schumacher (*brain*) and Amanda Montañez (*cascade diagram*)

Predictive processing may also help scientists understand developmental disorders such as autism. According to Nagai, certain autistic individuals may have a higher sensitivity to prediction errors, making incoming sensory information overwhelming. That could explain their attraction to repetitive behavior, whose outcomes are highly predictable.

Harold Bekkering, a cognitive psychologist at Radboud University in the Netherlands, believes that predictive processing could also help explain the behavior of people with attention deficit hyperactivity disorder. According to this theory, autistic individuals prefer to protect themselves from the unknown, whereas those who have trouble focusing are perpetually attracted to unpredictable stimuli in their surroundings, Bekkering explains. “Some people who are sensitive to the world explore the world, while other people who are too sensitive for the world shield themselves,” he suggests. “In a predictive coding framework, you can very nicely simulate both patterns.” His lab is currently working on using human brain imaging to test this hypothesis.

Nagai hopes to assess this theory by conducting “cognitive mirroring” studies in which robots, equipped with predictive learning algorithms, will interact with people. As the robot and person communicate using body language and facial expressions, the machine will adjust its behaviors to match its partner—thus reflecting the person's preference for predictability. In this way, experimenters can use robots to model human cognition—then examine its neural architecture to try to decipher what is going on inside human heads. “We can externalize our characteristics into robots to better understand ourselves,” Nagai says.

ROBOTS OF THE FUTURE

Studies of robotic children have thus helped answer certain key questions in psychology, including the importance of predictive processing, and of bodies, in cognitive development. “We have learned a huge amount about how complex systems work, how the body matters, [and] about really fundamental things like exploration and prediction,” Smith says.

Robots that can develop humanlike intelligence are far from becoming a reality, however: Chappie still belongs in the realm of science fiction. To begin with, scientists need to overcome technical hurdles, such as the brittle bodies and limited sensory capabilities of most robots. (Advances in areas such as soft robotics and robot vision may help this happen.) Far more challenging is the incredible intricacy of the brain itself. Despite efforts on many fronts to model the mind, scientists are far from engineering a machine to rival it. “I completely disagree with people who say that in 10 or 20 years we'll have machines with human-level intelligence,” Oudeyer says. “I think it's showing a profound misunderstanding of the complexity of human intelligence.”

Moreover, intelligence does not merely require the right machinery and circuitry. A long line of research has shown that caregivers are crucial to children's development. “If you ask me if a robot can become truly humanlike, then I'll ask you if somebody can take care of a robot like a child,” Tani says. “If that's possible, then yes, we might be able to do it, but otherwise, it's impossible to expect a robot to develop like a real human child.”

The process of gradually accumulating knowledge may also be indispensable. “Development is a very complex system of cascades,” Smith says. “What happens on one day lays the groundwork for [the next].” As a result, she argues, it might not be possible to build human-level artificial intelligence without somehow integrating the step-by-step process of learning that occurs throughout life.

Soon before his death, Richard Feynman famously wrote: “What I cannot create, I do not understand.” In Tani's 2016 book, *Exploring Robotic Minds*, he turns his concept around, saying, “I can understand what I can create.” The best way to understand the human mind, he argues, is to synthesize one.

One day humans may succeed in creating a robot that can explore, adapt and develop just like a child, perhaps complete with surrogate caregivers to provide the affection and guidance needed for healthy growth. In the meantime, childlike robots will continue to provide valuable insights into how children learn—and reveal what might happen when basic mechanisms go awry.

This article was originally published with the title "Self-Taught Robots"

MORE TO EXPLORE

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FROM OUR ARCHIVES

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Smithsonian.com

Will A.I. Ever Be Smarter Than a Four-Year-Old?

Looking at how children process information may give programmers useful hints about directions for computer learning

(FatCamera/iStock)

By Alison Gopnik
smithsonian.com
February 22, 2019

Everyone's heard about the new advances in artificial intelligence, and especially machine learning. You've also heard utopian or apocalyptic predictions about what those advances mean. They have been taken to presage either immortality or the end of the world, and a lot has been written about both of those possibilities. But the most sophisticated AIs are still far from being able to solve problems that human four-year-olds accomplish with ease. In spite of the impressive name, artificial intelligence largely consists of techniques to detect statistical patterns in large data sets. There is much more to human learning.

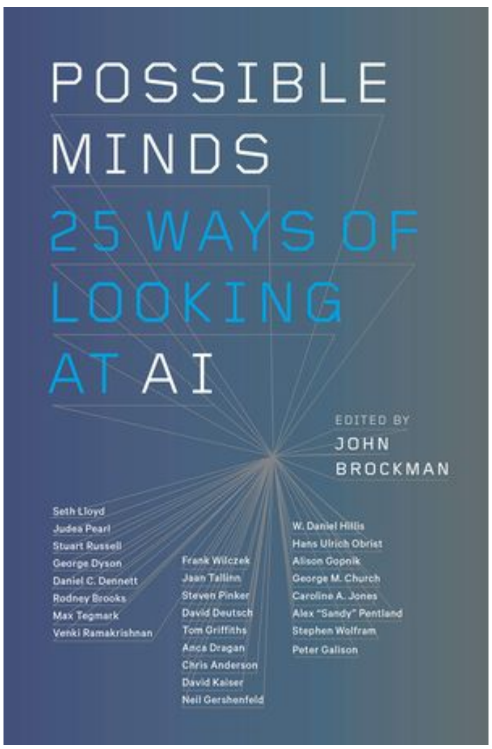
How can we possibly know so much about the world around us? We learn an enormous amount even when we are small children; four-year-olds already know about plants and animals and machines; desires, beliefs, and emotions; even dinosaurs and spaceships.

Science has extended our knowledge about the world to the unimaginably large and the infinitesimally small, to the edge of the universe and the beginning of time. And we use that knowledge to make new classifications and predictions, imagine new possibilities, and make new things happen in the world. But all that reaches any of us from the world is a stream of photons hitting our retinas and disturbances of air at our eardrums. How do we learn so much about the world when the evidence we have is so limited? And how do we do all this with the few pounds of grey goo that sits behind our eyes?

The best answer so far is that our brains perform computations on the concrete, particular, messy data arriving at our senses, and those computations yield accurate representations of the world. The representations seem to be structured, abstract, and hierarchical; they include the perception of three-dimensional objects, the grammars that underlie language, and mental capacities like "theory of mind," which lets us understand what other people think. Those representations allow us to make a wide range of new predictions and imagine many new possibilities in a distinctively creative human way.

This kind of learning isn't the only kind of intelligence, but it's a particularly important one for human beings. And it's the kind of intelligence that is a specialty of young children. Although children are dramatically bad at planning and decision making, they are the best learners in the universe. Much of the process of turning data into theories happens before we are five.

Since Aristotle and Plato, there have been two basic ways of addressing the problem of how we know what we know, and they are still the main approaches in machine learning. Aristotle approached the problem from the bottom up: Start with senses—the stream of photons and air vibrations (or the pixels or sound samples of a digital image or recording)—and see if you can extract patterns from them. This approach was carried further by such classic associationists as philosophers David Hume and J. S. Mill and later by behavioral psychologists, like Pavlov and B. F. Skinner. On this view, the abstractness and hierarchical structure of representations is something of an illusion, or at least an epiphenomenon. All the work can be done by association and pattern detection—especially if there are enough data.



Possible Minds: 25 Ways of Looking at AI

Science world luminary John Brockman assembles twenty-five of the most important scientific minds, people who have been thinking about the field artificial intelligence for most of their careers, for an unparalleled round-table examination about mind, thinking, intelligence and what it means to be human.

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Over time, there has been a seesaw between this bottom-up approach to the mystery of learning and Plato's alternative, top-down one. Maybe we get abstract knowledge from concrete data because we already know a lot, and especially because we already have an array of basic abstract concepts, thanks to evolution. Like scientists, we can use those concepts to formulate hypotheses about the world. Then, instead of trying to extract patterns from the raw data, we can make predictions about what the data should look like if those hypotheses are right. Along with Plato, such "rationalist" philosophers and psychologists as Descartes and Noam

Chomsky took this approach.

Here's an everyday example that illustrates the difference between the two methods: solving the spam plague. The data consist of a long, unsorted list of messages in your inbox. The reality is that some of these messages are genuine and some are spam. How can you use the data to discriminate between them?

Consider the bottom-up technique first. You notice that the spam messages tend to have particular features: a long list of addressees, origins in Nigeria, references to million-dollar prizes, or Viagra. The trouble is that perfectly useful messages might have these features, too. If you looked at enough examples of spam and nonspam emails, you might see not only that spam emails tend to have those features but that the features tend to go together in particular ways (Nigeria plus a million dollars spells trouble). In fact, there might be some subtle higher-level correlations that discriminate the spam messages from the useful ones—a particular pattern of misspellings and IP addresses, say. If you detect those patterns, you can filter out the spam.

The bottom-up machine-learning techniques do just this. The learner gets millions of examples, each with some set of features and each labeled as spam (or some other category) or not. The computer can extract the pattern of features that distinguishes the two, even if it's quite subtle.

How about the top-down approach? I get an email from the editor of the *Journal of Clinical Biology*. It refers to one of my papers and says that they would like to publish an article by me. No Nigeria, no Viagra, no million dollars; the email doesn't have any of the features of spam. But by using what I already know, and thinking in an abstract way about the process that produces spam, I can figure out that this email is suspicious:

1. I know that spammers try to extract money from people by appealing to human greed.

2. I also know that legitimate "open access" journals have started covering their costs by charging authors instead of subscribers, and that I don't practice anything like clinical biology.

Put all that together and I can produce a good new hypothesis about where that email came from. It's designed to sucker academics into paying to "publish" an article in a fake journal. The email was a result of the same dubious process as the other spam emails, even though it looked nothing like them. I can draw this conclusion from just one example, and I can go on to test my hypothesis further, beyond anything in the email itself, by googling the "editor."

In computer terms, I started out with a "generative model" that includes abstract concepts like greed and deception and describes the process that produces email scams. That lets me recognize the classic Nigerian email spam, but it also lets me imagine many different kinds of possible spam. When I get the journal email, I can work backward: "This seems like just the kind of mail that would come out of a spam-generating process."

The new excitement about AI comes because AI researchers have recently produced powerful and effective versions of both of these learning methods. But there is nothing profoundly new about the methods themselves.

Bottom-up Deep Learning

In the 1980s, computer scientists devised an ingenious way to get computers to detect patterns in data: connectionist, or neural-network, architecture (the "neural" part was, and still is, metaphorical). The approach fell into the doldrums in the 1990s but has recently been revived with powerful "deep-learning" methods like Google's DeepMind.

For example, you can give a deep-learning program a bunch of Internet images labeled "cat," others labeled "house," and so on. The program can detect the patterns differentiating the two sets of images and use that information to label new images correctly. Some kinds of machine learning, called unsupervised learning, can detect patterns in data with no labels at all; they simply look for clusters of features—what scientists call a factor analysis. In the deep-learning machines, these processes are repeated at different levels. Some programs can even discover relevant features from the raw data of pixels or sounds; the computer might begin by detecting the patterns in the raw image that correspond to edges and lines and then find the patterns in those patterns that correspond to faces, and so on.

Another bottom-up technique with a long history is reinforcement learning. In the 1950s, B. F. Skinner, building on the work of John Watson, famously programmed pigeons to perform elaborate actions—even guiding air-launched missiles to their targets (a disturbing echo of recent AI) by giving them a particular schedule of rewards and punishments. The essential idea was that actions that were rewarded would be repeated and those that were punished would not, until the desired behavior was achieved. Even in Skinner's day, this simple process, repeated over and over, could lead to complex behavior. Computers are designed to perform simple operations over and over on a scale that dwarfs human imagination, and computational systems can learn remarkably complex skills in this way.

For example, researchers at Google's DeepMind used a combination of deep learning and reinforcement learning to teach a computer to play Atari video games. The computer knew nothing about how the games worked. It began by acting randomly and got information only about what the screen looked like at each moment and how well it had scored. Deep learning helped interpret the features on the screen, and reinforcement learning rewarded the system for higher scores. The computer got very good at playing several of the games, but it also completely bombed on others that were just as easy for humans to master.

A similar combination of deep learning and reinforcement learning has enabled the success of DeepMind's AlphaZero, a program that managed to beat human players at both chess and Go, equipped with only a basic knowledge of the rules of the game and some planning capacities. AlphaZero has another interesting feature: It works by playing hundreds of millions of games against itself. As it does so, it prunes mistakes that led to losses, and it repeats and elaborates on strategies that led to wins. Such systems, and others involving techniques called generative adversarial networks, generate data as well as observing data.

When you have the computational power to apply those techniques to very large data sets or millions of email messages, Instagram images, or voice recordings, you can solve problems that seemed very difficult before. That's the source of much of the excitement in computer science. But it's worth remembering that those problems—like recognizing that an image is a cat or a spoken word is Siri—are trivial for a human toddler. One of the most interesting discoveries of computer science is that problems that are easy for us (like identifying cats) are hard for computers—much harder than playing chess or Go. Computers need millions of examples to categorize objects that we can categorize with just a few. These bottom-up systems can generalize to new examples; they can label a new image as a cat fairly accurately over all. But they do so in ways quite different from how humans generalize. Some images almost identical to a cat image won't be identified by us as cats at all. Others that look like a random blur will be.

Top-Down Bayesian Models

The top-down approach played a big role in early AI, and in the 2000s it, too, experienced a revival, in the form of probabilistic, or Bayesian, generative models.

The early attempts to use this approach faced two kinds of problems. First, most patterns of evidence might in principle be explained by many different hypotheses: It's possible that my journal email message is genuine, it just doesn't seem likely. Second, where do the concepts that the generative models use come from in the first place? Plato and Chomsky said you were born with them. But how can we explain how we learn the latest concepts of science? Or how even young children understand about dinosaurs and rocket ships?

Bayesian models combine generative models and hypothesis testing with probability theory, and they address these two problems. A Bayesian model lets you calculate just how likely it is that a particular hypothesis is true, given the data. And by making small but systematic tweaks to the models we already have, and testing them against the data, we can sometimes make new concepts and models from old ones. But these advantages are offset by other problems. The Bayesian techniques can help you choose which of two hypotheses is more likely, but there are almost always an enormous number of possible hypotheses, and no system can efficiently consider them all. How do you decide which hypotheses are worth testing in the first place?

Brenden Lake at NYU and colleagues have used these kinds of top-down methods to solve another problem that's easy for people but extremely difficult for computers: recognizing unfamiliar handwritten characters. Look at a character on a Japanese scroll. Even if you've never seen it before, you can probably tell if it's similar to or different from a character on another Japanese scroll. You can probably draw it and even design a fake Japanese character based on the one you see—one that will look quite different from a Korean or Russian character.

The bottom-up method for recognizing handwritten characters is to give the computer thousands of examples of each one and let it pull out the salient features. Instead, Lake et al. gave the program a general model of how you draw a character: A stroke goes either right or left; after you finish one, you start another; and so on. When the program saw a particular character, it could infer the sequence of strokes that were most likely to have led to it—just as I inferred that the spam pro- cess led to my dubious email. Then it could judge whether a new character was likely to result from that sequence or from a different one, and it could produce a similar set of strokes itself. The program worked much better than a deep-learning program applied to exactly the same data, and it closely mirrored the performance of human beings.

These two approaches to machine learning have complementary strengths and weaknesses. In the bottom-up approach, the program doesn't need much knowledge to begin with, but it needs a great deal of data, and it can generalize only in a limited way. In the top-down approach, the program can learn from just a few examples and make much broader and more varied generalizations, but you need to build much more into it to begin with. A number of investigators are currently trying to combine the two approaches, using deep learning to implement Bayesian inference.

The recent success of AI is partly the result of extensions of those old ideas. But it has more to do with the fact that, thanks to the Internet, we have much more data, and thanks to Moore's Law we have much more computational power to apply to that data. Moreover, an unappreciated fact is that the data we do have has already been sorted and processed by human beings. The cat pictures posted to the Web are canonical cat pictures—pictures that humans have already chosen as "good" pictures. Google Translate works because it takes advantage of millions of human translations and generalizes them to a new piece of text, rather than genuinely understanding the sentences themselves.

But the truly remarkable thing about human children is that they somehow combine the best features of each approach and then go way beyond them. Over the past fifteen years, developmentalists have been exploring the way children learn structure from data. Four-year-olds can learn by taking just one or two examples of data, as a top-down system does, and generalizing to very different concepts. But they can also learn new concepts and models from the data itself, as a bottom-up system does.

For example, in our lab we give young children a "blicket detector"—a new machine to figure out, one they've never seen before. It's a box that lights up and plays music when you put certain objects on it but not others. We give children just one or two examples of how the machine works, showing them that, say, two red blocks make it go, while a green-and-yellow combination doesn't. Even eighteen-month-olds immediately figure out the general principle that the two objects have to be the same to make it go, and they generalize that principle to new examples: For instance, they will choose two objects that have the same *shape* to make the machine work. In other experiments, we've shown that children can even figure out that some hidden invisible property makes the machine go, or that the machine works on some abstract logical principle.

You can show this in children's everyday learning, too. Young children rapidly learn abstract intuitive theories of biology, physics, and psychology in much the way adult scientists do, even with relatively little data.

The remarkable machine-learning accomplishments of the recent AI systems, both bottom-up and top-down, take place in a narrow and well-defined space of hypotheses and concepts—a precise set of game pieces and moves, a predetermined set of images. In contrast, children and scientists alike sometimes change their concepts in radical ways, performing paradigm shifts rather than simply tweaking the concepts they already have.

Four-year-olds can immediately recognize cats and understand words, but they can also make creative and surprising new inferences that go far beyond their experience. My own grandson recently explained, for example, that if an adult wants to become a child again, he should try not eating any healthy vegetables, since healthy vegetables make a child grow into an adult. This kind of hypothesis, a plausible one that no grown-up would ever entertain, is characteristic of young children. In fact, my colleagues and I have shown systematically that preschoolers are better at coming up with unlikely hypotheses than older children and adults. We have almost no idea how this kind of creative learning and innovation is possible.

Looking at what children do, though, may give programmers useful hints about directions for computer learning. Two features of children's learning are especially striking. Children are active learners; they don't just passively soak up data like AIs do. Just as scientists experiment, children are intrinsically motivated to extract information from the world around them through their endless play and exploration. Recent studies show that this exploration is more systematic than it looks and is well adapted to find persuasive evidence to support hypothesis formation and theory choice. Building curiosity into machines and allowing them to actively interact with the world might be a route to more realistic and wide-ranging learning.

Second, children, unlike existing AIs, are social and cultural learners. Humans don't learn in isolation but avail themselves of the accumulated wisdom of past generations. Recent studies show that even preschoolers learn through imitation and by listening to the testimony of others. But they don't simply passively obey their teachers. Instead they take in information from others in a remarkably subtle and sensitive way, making complex inferences about where the information comes from and how trustworthy it is and systematically integrating their own experiences with what they are hearing.

"Artificial intelligence" and "machine learning" sound scary. And in some ways they are. These systems are being used to control weapons, for example, and we really should be scared about that. Still, natural stupidity can wreak far more havoc than artificial intelligence; we humans will need to be much smarter than we have been in the past to properly regulate the new technologies. But there is not much basis for either the apocalyptic or the utopian vision of AIs replacing humans. Until we solve the basic paradox of learning, the best artificial intelligences will be unable to compete with the average human four-year-old.